

# 11

## Structured Singular Value

It is noted that the robust stability and robust performance criteria derived in Chapter 9 vary with the assumptions on the uncertainty descriptions and performance requirements. We will show in this chapter that they can all be treated in a unified framework using the LFT machinery introduced in the last chapter and the structured singular value to be introduced in this chapter. This, of course, does not mean that those special problems and their corresponding results are not important; on the contrary, they are sometimes very enlightening to our understanding of complex problems such as those in which complex problems are formed from simple problems. On the other hand, a unified approach may relieve the mathematical burden of dealing with specific problems repeatedly. Furthermore, the unified framework introduced here will enable us to treat exactly the robust stability and robust performance problems for systems with multiple sources of uncertainties, which is a formidable problem in the standing point of Chapter 9, in the same fashion as single unstructured uncertainty. Indeed, if a system is subject to multiple sources of uncertainties, in order to use the results in Chapter 9 for unstructured cases, it is necessary to reflect all sources of uncertainties from their known point of occurrence to a single reference location in the loop. Such reflected uncertainties invariably have a great deal of structure which must then be “covered up” with a large, arbitrarily more conservative perturbation in order to maintain a simple cone bounded representation at the reference location. Readers might have already had some idea about the conservativeness in such reflection from the skewed specification problem, where an input multiplicative uncertainty of the plant is reflected at the output and the size of the reflected uncertainty is proportional to the condition number of the plant. In general, the reflected uncertainty may be proportional to the condition

number of the transfer matrix between its original location and the reflected location. Thus it is highly desirable to treat the uncertainties as they are and where they are. The structured singular value is defined exactly for that purpose.

## 11.1 General Framework for System Robustness

As we have illustrated in the last chapter, any interconnected system may be rearranged to fit the general framework in Figure 11.1. Although the interconnection structure can become quite complicated for complex systems, many software packages, such as SIMULINK<sup>1</sup> and  $\mu$ -TOOLS<sup>2</sup>, are available which could be used to generate the interconnection structure from system components. Various modeling assumptions will be considered and the impact of these assumptions on analysis and synthesis methods will be explored in this general framework.

Note that uncertainty may be modeled in two ways, either as external inputs or as perturbations to the nominal model. The performance of a system is measured in terms of the behavior of the outputs or errors. The assumptions which characterize the uncertainty, performance, and nominal models determine the analysis techniques which must be used. The models are assumed to be FDLTI systems. The uncertain inputs are assumed to be either filtered white noise or weighted power or weighted  $\mathcal{L}_p$  signals. Performance is measured as weighted output variances, or as power, or as weighted output  $\mathcal{L}_p$  norms. The perturbations are assumed to be themselves FDLTI systems which are norm-bounded as input-output operators. Various combinations of these assumptions form the basis for all the standard linear system analysis tools.

Given that the nominal model is an FDLTI system, the interconnection system has the form

$$P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) & P_{13}(s) \\ P_{21}(s) & P_{22}(s) & P_{23}(s) \\ P_{31}(s) & P_{32}(s) & P_{33}(s) \end{bmatrix}$$

and the closed-loop system is an LFT on the perturbation and the controller given by

$$\begin{aligned} z &= \mathcal{F}_u(\mathcal{F}_\ell(P, K), \Delta) w \\ &= \mathcal{F}_\ell(\mathcal{F}_u(P, \Delta), K) w. \end{aligned}$$

We will focus our discussion in this section on analysis methods; therefore, the controller may be viewed as just another system component and absorbed into the interconnection structure. Denote

$$M(s) = \mathcal{F}_\ell(P(s), K(s)) = \begin{bmatrix} M_{11}(s) & M_{12}(s) \\ M_{21}(s) & M_{22}(s) \end{bmatrix},$$

<sup>1</sup>SIMULINK is a trademark of The MathWorks, Inc.

<sup>2</sup> $\mu$ -TOOLS is a trademark of MUSYN Inc.

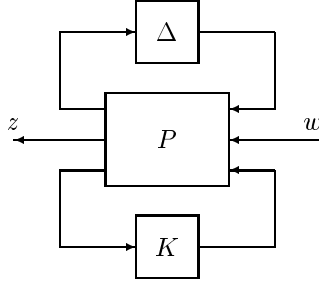


Figure 11.1: General Framework

and then the general framework reduces to Figure 11.2, where

$$z = \mathcal{F}_u(M, \Delta)w = [M_{22} + M_{21}\Delta(I - M_{11}\Delta)^{-1}M_{12}] w.$$

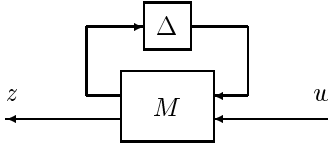


Figure 11.2: Analysis Framework

Suppose  $K(s)$  is a stabilizing controller for the nominal plant  $P$ . Then  $M(s) \in \mathcal{RH}_\infty$ . In general, the stability of  $\mathcal{F}_u(M, \Delta)$  does not necessarily imply the internal stability of the closed-loop feedback system. However, they can be made equivalent with suitably chosen  $w$  and  $z$ . For example, consider again the multiplicatively perturbed system shown in Figure 11.3. Now let

$$w := \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}, \quad z := \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

then the system is robustly stable for all  $\Delta(s) \in \mathcal{RH}_\infty$  with  $\|\Delta\|_\infty < 1$  if and only if  $\mathcal{F}_u(M, \Delta) \in \mathcal{RH}_\infty$  for all admissible  $\Delta$ , which is guaranteed by  $\|M_{11}\|_\infty \leq 1$ . (Note that this is not necessarily equivalent to  $(I - M_{11}\Delta)^{-1} \in \mathcal{RH}_\infty$  if  $\Delta$  belongs to a closed ball as shown in Theorem 9.5.)

The analysis results presented in the previous chapters together with the associated synthesis tools are summarized in Table 11.1 with various uncertainty modeling assumptions.

However, the analysis is not so simple for systems with multiple sources of model uncertainties, including the robust performance problem for systems with unstructured

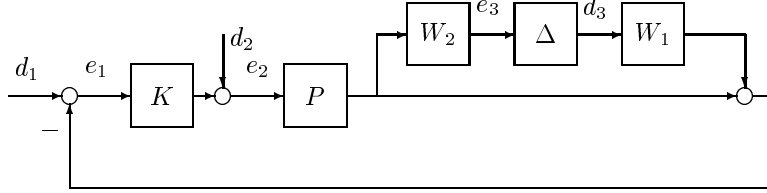


Figure 11.3: Multiplicatively Perturbed Systems

uncertainty. As we have shown in the last chapter, if a system is built from components which are themselves uncertain, then, in general, the uncertainty in the system level is structured involving typically a large number of real parameters. The stability analysis involving real parameters is much more involved and is beyond the scope of this book and that, instead, we shall simply cover the real parametric uncertainty with norm bounded dynamical uncertainty. Moreover, the interconnection model  $M$  can always be chosen so that  $\Delta(s)$  is block diagonal, and, by absorbing any weights,  $\|\Delta\|_\infty < 1$ . Thus we shall assume that  $\Delta(s)$  takes the form of

$$\Delta(s) = \{\text{diag} [\delta_1 I_{r_1}, \dots, \delta_s I_{r_s}, \Delta_1, \dots, \Delta_F] : \delta_i(s) \in \mathcal{RH}_\infty, \Delta_j \in \mathcal{RH}_\infty\}$$

with  $\|\delta_i\|_\infty < 1$  and  $\|\Delta_j\|_\infty < 1$ . Then the system is robustly stable iff the interconnected system in Figure 11.4 is stable.

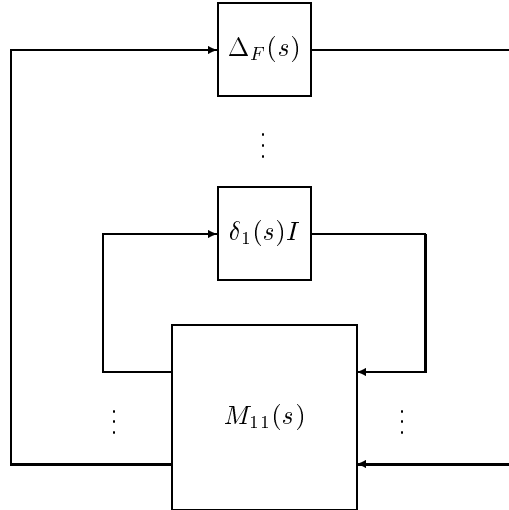


Figure 11.4: Robust Stability Analysis Framework

The results of Table 11.1 can be applied to the analyses of the system's robust stability in two ways:

| Input Assumptions                       | Performance Specifications | Perturbation Assumptions | Analysis Tests             | Synthesis Methods              |
|-----------------------------------------|----------------------------|--------------------------|----------------------------|--------------------------------|
| $E(w(t)w(t)^*) = I$                     | $E(z(t)^*z(t)) \leq 1$     | $\Delta = 0$             | $\ M_{22}\ _2 \leq 1$      | LQG                            |
| $w = U_0\delta(t)$<br>$E(U_0U_0^*) = I$ | $E(\ z\ _2^2) \leq 1$      |                          |                            | Wiener-Hopf<br>$\mathcal{H}_2$ |
| $\ w\ _2 \leq 1$                        | $\ z\ _2 \leq 1$           | $\Delta = 0$             | $\ M_{22}\ _\infty \leq 1$ | Singular Value Loop Shaping    |
| $\ w\ _2 \leq 1$                        | Internal Stability         | $\ \Delta\ _\infty < 1$  | $\ M_{11}\ _\infty \leq 1$ |                                |

Table 11.1: General Analysis for Single Source of Uncertainty

- (1)  $\|M_{11}\|_\infty \leq 1$  implies stability, but not conversely, because this test ignores the known block diagonal structure of the uncertainties and is equivalent to regarding  $\Delta$  as unstructured. This can be arbitrarily conservative<sup>3</sup> in that stable systems can have arbitrarily large  $\|M_{11}\|_\infty$ .
- (2) Test for each  $\delta_i$  ( $\Delta_j$ ) individually (assuming no uncertainty in other channels). This test can be arbitrarily optimistic because it ignores interaction between the  $\delta_i$  ( $\Delta_j$ ). This optimism is also clearly shown in the spinning body example.

The difference between the stability margins(or bounds on  $\Delta$ ) obtained in (1) and (2) can be arbitrarily far apart. Only when the margins are close can conclusions be made about the general case with structured uncertainty.

The exact stability and performance analysis for systems with structured uncertainty requires a new matrix function called the structured singular value (SSV) which is denoted by  $\mu$ .

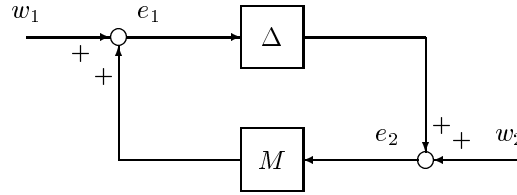
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<sup>3</sup>By “arbitrarily conservative,” we mean that examples can be constructed where the degree of conservatism is arbitrarily large. Of course, other examples exist where it is quite reasonable, see for example the spinning body example.

## 11.2 Structured Singular Value

### 11.2.1 Basic Concept

Conceptually, the structured singular value is nothing but a straightforward generalization of the singular values for constant matrices. To be more specific, it is instructive at this point to consider again the robust stability problem of the following standard feedback interconnection with stable  $M(s)$  and  $\Delta(s)$ .



One important question one might ask is how large  $\Delta$  (in the sense of  $\|\Delta\|_\infty$ ) can be without destabilizing the feedback system. Since the closed-loop poles are given by  $\det(I - M\Delta) = 0$ , the feedback system becomes unstable if  $\det(I - M(s)\Delta(s)) = 0$  for some  $s \in \overline{\mathbb{C}}_+$ . Now let  $\alpha > 0$  be a sufficiently small number such that the closed-loop system is stable for all stable  $\|\Delta\|_\infty < \alpha$ . Next increase  $\alpha$  until  $\alpha_{max}$  so that the closed-loop system becomes unstable. So  $\alpha_{max}$  is the robust stability margin. By small gain theorem,

$$\frac{1}{\alpha_{max}} = \|M\|_\infty := \sup_{s \in \overline{\mathbb{C}}_+} \overline{\sigma}(M(s)) = \sup_{\omega} \overline{\sigma}(M(j\omega))$$

if  $\Delta$  is unstructured. Note that for any fixed  $s \in \overline{\mathbb{C}}_+$ ,  $\overline{\sigma}(M(s))$  can be written as

$$\overline{\sigma}(M(s)) = \frac{1}{\min \{ \overline{\sigma}(\Delta) : \det(I - M(s)\Delta) = 0, \Delta \text{ is unstructured} \}}. \quad (11.1)$$

In other words, the reciprocal of the *largest singular value* of  $M$  is a measure of the smallest unstructured  $\Delta$  that causes instability of the feedback system.

To quantify the smallest destabilizing *structured*  $\Delta$ , the concept of singular values needs to be generalized. In view of the characterization of the largest singular value of a matrix  $M(s)$  given by (11.1), we shall define

$$\mu_\Delta(M(s)) = \frac{1}{\min \{ \overline{\sigma}(\Delta) : \det(I - M(s)\Delta) = 0, \Delta \text{ is structured} \}} \quad (11.2)$$

as the largest structured singular value of  $M(s)$  with respect to the structured  $\Delta$ . Then it is obvious that the robust stability margin of the feedback system with structured uncertainty  $\Delta$  is

$$\frac{1}{\alpha_{max}} = \sup_{s \in \overline{\mathbb{C}}_+} \mu_\Delta(M(s)) = \sup_{\omega} \mu_\Delta(M(j\omega)).$$

The last equality follows from the following lemma.

**Lemma 11.1** *Let  $\Delta$  be a structured set and  $M(s) \in \mathcal{RH}_\infty$ . Then*

$$\sup_{s \in \overline{\mathbb{C}}_+} \mu_\Delta(M(s)) = \sup_{s \in \mathbb{C}_+} \mu_\Delta(M(s)) = \sup_{\omega} \mu_\Delta(M(j\omega)).$$

**Proof.** It is clear that

$$\sup_{s \in \overline{\mathbb{C}}_+} \mu_\Delta(M(s)) = \sup_{s \in \mathbb{C}_+} \mu_\Delta(M(s)) \geq \sup_{\omega} \mu_\Delta(M(j\omega)).$$

Now suppose  $\sup_{s \in \mathbb{C}_+} \mu_\Delta(M(s)) > 1/\alpha$ , then by the definition of  $\mu$ , there is an  $s_o \in \overline{\mathbb{C}}_+ \cup \{\infty\}$  and a complex structured  $\Delta$  such that  $\overline{\sigma}(\Delta) < \alpha$  and  $\det(I - M(s_o)\Delta) = 0$ . This implies that there is a  $0 < \hat{\omega} \leq \infty$  and  $0 < \beta \leq 1$  such that  $\det(I - M(j\hat{\omega})\beta\Delta) = 0$ . This in turn implies that  $\mu_\Delta(M(j\hat{\omega})) > 1/\alpha$  since  $\overline{\sigma}(\beta\Delta) < \alpha$ . In other words,  $\sup_{s \in \mathbb{C}_+} \mu_\Delta(M(s)) \leq \sup_{\omega} \mu_\Delta(M(j\omega))$ . The proof is complete.  $\square$

The formal definition and characterization of the structured singular value of a constant matrix will be given below.

### 11.2.2 Definitions of $\mu$

This section is devoted to defining the structured singular value, a matrix function denoted by  $\mu(\cdot)$ . We consider matrices  $M \in \mathbb{C}^{n \times n}$ . In the definition of  $\mu(M)$ , there is an underlying structure  $\Delta$  (a prescribed set of block diagonal matrices) on which everything in the sequel depends. For each problem, this structure is, in general, different; it depends on the *uncertainty* and *performance objectives* of the problem. Defining the structure involves specifying three things: the type of each block, the total number of blocks, and their dimensions.

There are two types of blocks: *repeated scalar* and *full* blocks. Two nonnegative integers,  $S$  and  $F$ , represent the number of *repeated scalar* blocks and the number of *full* blocks, respectively. To bookkeep their dimensions, we introduce positive integers  $r_1, \dots, r_S; m_1, \dots, m_F$ . The  $i$ 'th repeated scalar block is  $r_i \times r_i$ , while the  $j$ 'th full block is  $m_j \times m_j$ . With those integers given, we define  $\Delta \subset \mathbb{C}^{n \times n}$  as

$$\Delta = \{\text{diag} [\delta_1 I_{r_1}, \dots, \delta_S I_{r_S}, \Delta_1, \dots, \Delta_F] : \delta_i \in \mathbb{C}, \Delta_j \in \mathbb{C}^{m_j \times m_j}\}. \quad (11.3)$$

For consistency among all the dimensions, we must have

$$\sum_{i=1}^S r_i + \sum_{j=1}^F m_j = n.$$

Often, we will need norm bounded subsets of  $\Delta$ , and we introduce the following notation:

$$\mathbf{B}\Delta = \{\Delta \in \Delta : \overline{\sigma}(\Delta) \leq 1\} \quad (11.4)$$

$$\mathbf{B}^\circ \Delta = \{\Delta \in \Delta : \overline{\sigma}(\Delta) < 1\} \quad (11.5)$$

where the superscript “o” symbolizes the open ball. To keep the notation as simple as possible in (11.3), we place all of the repeated scalar blocks first; in actuality, they can come in any order. Also, the full blocks do not have to be square, but restricting them as such saves a great deal in terms of notation.

**Definition 11.1** For  $M \in \mathbb{C}^{n \times n}$ ,  $\mu_{\Delta}(M)$  is defined as

$$\mu_{\Delta}(M) := \frac{1}{\min \{\overline{\sigma}(\Delta) : \Delta \in \Delta, \det(I - M\Delta) = 0\}} \quad (11.6)$$

unless no  $\Delta \in \Delta$  makes  $I - M\Delta$  singular, in which case  $\mu_{\Delta}(M) := 0$ .

**Remark 11.1** Without a loss in generality, the full blocks in the minimal norm  $\Delta$  can each be chosen to be dyads (rank = 1). To see this, assume  $S = 0$ , i.e., all blocks are full blocks. Suppose that  $I - M\Delta$  is singular for some  $\Delta \in \Delta$ . Then there is an  $x \in \mathbb{C}^n$  such that  $M\Delta x = x$ . Now partition  $x$  conformably with  $\Delta$ :

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_F \end{bmatrix}, \quad x_i \in \mathbb{C}^{m_i}, i = 1, \dots, F$$

and define

$$D = \text{diag}(d_1 I_{m_1}, \dots, d_F I_{m_F})$$

where  $d_i = 1$  if  $x_i = 0$  and  $d_i = 1/\|x_i\|$  if  $x_i \neq 0$ . Next define

$$\tilde{x} = \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_F \end{bmatrix} := Dx = \begin{bmatrix} d_1 x_1 \\ d_2 x_2 \\ \vdots \\ d_F x_F \end{bmatrix}$$

and

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_F \end{bmatrix} := \Delta \tilde{x} = \begin{bmatrix} \Delta_1 \tilde{x}_1 \\ \Delta_2 \tilde{x}_2 \\ \vdots \\ \Delta \tilde{x}_F \end{bmatrix}.$$

It follows that  $\|\tilde{x}_i\| = 1$  if  $x_i \neq 0$ ,  $\|\tilde{x}_i\| = 0$  if  $x_i = 0$ , and  $y \neq 0$ . Hence, define a new perturbation  $\tilde{\Delta} \in \mathbb{C}^{n \times n}$  as

$$\tilde{\Delta} := \text{diag} [y_1 \tilde{x}_1^*, \dots, y_F \tilde{x}_F^*].$$



Obviously,  $\bar{\sigma}(\tilde{\Delta}) \leq \bar{\sigma}(\Delta)$  and  $y = \tilde{\Delta}\tilde{x}$ . Note that  $D\Delta = \Delta D$  and  $D\tilde{\Delta} = \tilde{\Delta}D$ , we have

$$\begin{aligned} M\Delta x = x &\implies MD^{-1}\Delta\tilde{x} = x \implies MD^{-1}y = x \\ &\implies MD^{-1}\tilde{\Delta}\tilde{x} = x \implies M\tilde{\Delta}x = x \end{aligned}$$

i.e.,  $I - M\tilde{\Delta}$  is also singular. Hence we have replaced a general perturbation  $\Delta$  which satisfies the singularity condition with a perturbation  $\tilde{\Delta}$  that is no larger (in the  $\bar{\sigma}(\cdot)$  sense) and has rank 1 for each blocks but still satisfies the singularity condition.  $\heartsuit$

An alternative expression for  $\mu_{\Delta}(M)$  follows from the definition.

**Lemma 11.2**  $\mu_{\Delta}(M) = \max_{\Delta \in \mathbf{B}_{\Delta}} \rho(M\Delta)$

In view of this lemma, continuity of the function  $\mu: \mathbb{C}^{n \times n} \rightarrow \mathbb{R}$  is apparent. In general, though, the function  $\mu: \mathbb{C}^{n \times n} \rightarrow \mathbb{R}$  is not a norm, since it doesn't satisfy the triangle inequality; however, for any  $\alpha \in \mathbb{C}$ ,  $\mu(\alpha M) = |\alpha|\mu(M)$ , so in some sense, it is related to how "big" the matrix is.

We can relate  $\mu_{\Delta}(M)$  to familiar linear algebra quantities when  $\Delta$  is one of two extreme sets.

- If  $\Delta = \{\delta I : \delta \in \mathbb{C}\}$  ( $S=1, F=0, r_1=n$ ), then  $\mu_{\Delta}(M) = \rho(M)$ , the spectral radius of  $M$ .

**Proof.** The only  $\Delta$ 's in  $\Delta$  which satisfy the  $\det(I - M\Delta) = 0$  constraint are reciprocals of nonzero eigenvalues of  $M$ . The smallest one of these is associated with the largest (magnitude) eigenvalue, so,  $\mu_{\Delta}(M) = \rho(M)$ .  $\square$

- If  $\Delta = \mathbb{C}^{n \times n}$  ( $S=0, F=1, m_1=n$ ), then  $\mu_{\Delta}(M) = \bar{\sigma}(M)$ .

**Proof.** If  $\bar{\sigma}(\Delta) < \frac{1}{\bar{\sigma}(M)}$ , then  $\bar{\sigma}(M\Delta) < 1$ , so  $I - M\Delta$  is nonsingular. Applying equation (11.6) implies  $\mu_{\Delta}(M) \leq \bar{\sigma}(M)$ . On the other hand, let  $u$  and  $v$  be unit vectors satisfying  $Mv = \bar{\sigma}(M)u$ , and define  $\Delta := \frac{1}{\bar{\sigma}(M)}vu^*$ . Then  $\bar{\sigma}(\Delta) = \frac{1}{\bar{\sigma}(M)}$  and  $I - M\Delta$  is obviously singular. Hence,  $\mu_{\Delta}(M) \geq \bar{\sigma}(M)$ .  $\square$

Obviously, for a general  $\Delta$  as in (11.3) we must have

$$\{\delta I_n : \delta \in \mathbb{C}\} \subset \Delta \subset \mathbb{C}^{n \times n}. \quad (11.7)$$

Hence directly from the definition of  $\mu$  and from the two special cases above, we conclude that

$$\rho(M) \leq \mu_{\Delta}(M) \leq \bar{\sigma}(M). \quad (11.8)$$

These bounds alone are not sufficient for our purposes because the gap between  $\rho$  and  $\bar{\sigma}$  can be arbitrarily large. For example, suppose  $\Delta = \begin{bmatrix} \delta_1 & 0 \\ 0 & \delta_2 \end{bmatrix}$  and consider

- (1)  $M = \begin{bmatrix} 0 & \beta \\ 0 & 0 \end{bmatrix}$  for any  $\beta > 0$ . Then  $\rho(M) = 0$  and  $\bar{\sigma}(M) = \beta$ . But  $\det(I - M\Delta) = 1$  so  $\mu(M) = 0$ .
- (2)  $M = \begin{bmatrix} -1/2 & 1/2 \\ -1/2 & 1/2 \end{bmatrix}$ . Then  $\rho(M) = 0$  and  $\bar{\sigma}(M) = 1$ . Since  $\det(I - M\Delta) = 1 + \frac{\delta_1 - \delta_2}{2}$ , it is easy to see that  $\min \{ \max_i |\delta_i|, 1 + \frac{\delta_1 - \delta_2}{2} = 0 \} = 1$ , so  $\mu(M) = 1$ .

Thus neither  $\rho$  nor  $\bar{\sigma}$  provide useful bounds even in simple cases. The only time they do provide reliable bounds is when  $\rho \approx \bar{\sigma}$ .

However, the bounds can be refined by considering transformations on  $M$  that *do not affect*  $\mu_{\Delta}(M)$ , but *do affect*  $\rho$  and  $\bar{\sigma}$ . To do this, define the following two subsets of  $\mathbb{C}^{n \times n}$ :

$$\mathcal{U} = \{U \in \Delta : UU^* = I_n\} \quad (11.9)$$

$$\mathcal{D} = \left\{ \begin{array}{l} \text{diag} [D_1, \dots, D_S, d_1 I_{m_1}, \dots, d_{F-1} I_{m_{F-1}}, I_{m_F}] : \\ D_i \in \mathbb{C}^{r_i \times r_i}, D_i = D_i^*, d_j \in \mathbb{R}, d_j > 0 \end{array} \right\}. \quad (11.10)$$

Note that for any  $\Delta \in \Delta, U \in \mathcal{U}$ , and  $D \in \mathcal{D}$ ,

$$U^* \in \mathcal{U} \quad U\Delta \in \Delta \quad \Delta U \in \Delta \quad \bar{\sigma}(U\Delta) = \bar{\sigma}(\Delta U) = \bar{\sigma}(\Delta) \quad (11.11)$$

$$D\Delta = \Delta D. \quad (11.12)$$

Consequently,

**Theorem 11.3** For all  $U \in \mathcal{U}$  and  $D \in \mathcal{D}$

$$\mu_{\Delta}(MU) = \mu_{\Delta}(UM) = \mu_{\Delta}(M) = \mu_{\Delta}(DMD^{-1}). \quad (11.13)$$

**Proof.** For all  $D \in \mathcal{D}$  and  $\Delta \in \Delta$ ,

$$\det(I - M\Delta) = \det(I - MD^{-1}\Delta D) = \det(I - DMD^{-1}\Delta)$$

since  $D$  commutes with  $\Delta$ . Therefore  $\mu_{\Delta}(M) = \mu_{\Delta}(DMD^{-1})$ . Also, for each  $U \in \mathcal{U}$ ,  $\det(I - M\Delta) = 0$  if and only if  $\det(I - MUU^*\Delta) = 0$ . Since  $U^*\Delta \in \Delta$  and  $\bar{\sigma}(U^*\Delta) = \bar{\sigma}(\Delta)$ , we get  $\mu_{\Delta}(MU) = \mu_{\Delta}(M)$  as desired. The argument for  $UM$  is the same.  $\square$

Therefore, the bounds in (11.8) can be tightened to

$$\max_{U \in \mathcal{U}} \rho(UM) \leq \max_{\Delta \in \mathbf{B}\Delta} \rho(\Delta M) = \mu_{\Delta}(M) \leq \inf_{D \in \mathcal{D}} \bar{\sigma}(DMD^{-1}) \quad (11.14)$$

where the equality comes from Lemma 11.2. Note that the last element in the  $D$  matrix is normalized to 1 since for any nonzero scalar  $\gamma$ ,  $DMD^{-1} = (\gamma D)M(\gamma D)^{-1}$ .

**Remark 11.2** Note that the scaling set  $\mathcal{D}$  in Theorem 11.3 and in the inequality (11.14) does not necessarily be restricted to Hermitian. In fact, they can be replaced by any set of nonsingular matrices that satisfy (11.12). However, enlarging the set of scaling matrices does not improve the upper bound in inequality (11.14). This can be shown as follows: Let  $D$  be any nonsingular matrix such that  $D\Delta = \Delta D$ . Then there exist a Hermitian matrix  $0 < R = R^* \in \mathcal{D}$  and a unitary matrix  $U$  such that  $D = UR$  and

$$\inf_D \bar{\sigma}(DMD^{-1}) = \inf_D \bar{\sigma}(URMR^{-1}U^*) = \inf_{R \in \mathcal{D}} \bar{\sigma}(RMR^{-1}).$$

Therefore, there is no loss of generality in assuming  $\mathcal{D}$  to be Hermitian. ♡

### 11.2.3 Bounds

In this section we will concentrate on the bounds

$$\max_{U \in \mathcal{U}} \rho(UM) \leq \mu_{\Delta}(M) \leq \inf_{D \in \mathcal{D}} \bar{\sigma}(DMD^{-1}).$$

The lower bound is always an equality [Doyle, 1982].

**Theorem 11.4**  $\max_{U \in \mathcal{U}} \rho(MU) = \mu(M)$ .

Unfortunately, the quantity  $\rho(UM)$  can have multiple local maxima which are not global. Thus local search cannot be guaranteed to obtain  $\mu$ , but can only yield a lower bound. For computation purposes one can derive a slightly different formulation of the lower bound as a power algorithm which is reminiscent of power algorithms for eigenvalues and singular values [Packard, Fan, and Doyle, 1988]). While there are open questions about convergence, the algorithm usually works quite well and has proven to be an effective method to compute  $\mu$ .

The upper bound can be reformulated as a convex optimization problem, so the global minimum can, in principle, be found. Unfortunately, the upper bound is not always equal to  $\mu$ . For block structures  $\Delta$  satisfying  $2S + F \leq 3$ , the upper bound is always equal to  $\mu_{\Delta}(M)$ , and for block structures with  $2S + F > 3$ , there exist matrices for which  $\mu$  is less than the infimum. This can be summarized in the following diagram, which shows for which cases the upper bound is guaranteed to be equal to  $\mu$ .

**Theorem 11.5**  $\mu(M) = \inf_{D \in \mathcal{D}} \bar{\sigma}(DMD^{-1})$  if  $2S + F \leq 3$

| F= | 0   | 1   | 2   | 3   | 4  |
|----|-----|-----|-----|-----|----|
| S= |     |     |     |     |    |
| 0  |     | yes | yes | yes | no |
| 1  | yes | yes | no  | no  | no |
| 2  | no  | no  | no  | no  | no |

Several of the boxes have connections with standard results.

- $S = 0, F = 1$  :  $\mu_{\Delta}(M) = \bar{\sigma}(M)$ .
- $S = 1, F = 0$  :  $\mu_{\Delta}(M) = \rho(M) = \inf_{D \in \mathcal{D}} \bar{\sigma}(DM D^{-1})$ . This is a standard result in linear algebra. In fact, without a loss in generality, the matrix  $M$  can be assumed in Jordan Canonical form. Now let

$$J_1 = \begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{bmatrix}, \quad D_1 = \begin{bmatrix} 1 & & & & \\ & k & & & \\ & & \ddots & & \\ & & & k^{n_1-2} & \\ & & & & k^{n_1-1} \end{bmatrix} \in \mathbb{C}^{n_1 \times n_1}.$$

Then  $\inf_{D_1 \in \mathbb{C}^{n_1 \times n_1}} \bar{\sigma}(D_1 J_1 D_1^{-1}) = \lim_{k \rightarrow \infty} \bar{\sigma}(D_1 J_1 D_1^{-1}) = |\lambda|$ . (Note that by Remark 11.2, the scaling matrix does not need to be Hermitian.) The conclusion follows by applying this result to each Jordan block.

It is also equivalent to the fact that Lyapunov asymptotic stability and exponential stability are equivalent for discrete time systems. This is because  $\rho(M) < 1$  (exponential stability of a discrete time system matrix  $M$ ) implies for some non-singular  $D \in \mathbb{C}^{n \times n}$

$$\bar{\sigma}(DM D^{-1}) < 1 \quad \text{or} \quad (D^{-1})^* M^* D^* D M D^{-1} - I < 0$$

which in turn is equivalent to the existence of a  $P = D^* D > 0$  such that

$$M^* P M - P < 0$$

(Lyapunov asymptotic stability).

- $S = 0, F = 2$  : This case was studied by Redheffer [1959].
- $S = 1, F = 1$  : This is equivalent to a state space characterization of the  $\mathcal{H}_{\infty}$  norm of a discrete time transfer function, see Chapter 21.
- $S = 2, F = 0$  : This is equivalent to the fact that for multidimensional systems (2-d, in fact), exponential stability is not equivalent to Lyapunov stability.
- $S = 0, F \geq 4$  : For this case, the upper bound is not always equal to  $\mu$ . This is important, as these are the cases that arise most frequently in applications. Fortunately, the bound seems to be close to  $\mu$ . The worst known example has a ratio of  $\mu$  over the bound of about .85, and most systems are close to 1.

The above bounds are much more than just computational schemes. They are also theoretically rich and can unify a number of apparently quite different results in linear systems theory. There are several connections with Lyapunov asymptotic stability, two of which were hinted at above, but there are further connections between the upper bound scalings and solutions to Lyapunov and Riccati equations. Indeed, many major theorems in linear systems theory follow from the upper bounds and from some results of Linear Fractional Transformations. The lower bound can be viewed as a natural generalization of the maximum modulus theorem.

Of course one of the most important uses of the upper bound is as a computational scheme when combined with the lower bound. For reliable use of the  $\mu$  theory, it is essential to have upper and lower bounds. Another important feature of the upper bound is that it can be combined with  $H_\infty$  controller synthesis methods to yield an ad-hoc  $\mu$ -synthesis method. Note that the upper bound when applied to transfer functions is simply a scaled  $H_\infty$  norm. This is exploited in the  $D-K$  iteration procedure to perform approximate  $\mu$ -synthesis (Doyle[1982]), which will be briefly introduced in section 11.4.

#### 11.2.4 Well Posedness and Performance for Constant LFTs

Let  $M$  be a complex matrix partitioned as

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \quad (11.15)$$

and suppose there are two defined block structures,  $\Delta_1$  and  $\Delta_2$ , which are compatible in size with  $M_{11}$  and  $M_{22}$ , respectively. Define a third structure  $\Delta$  as

$$\Delta = \left\{ \begin{bmatrix} \Delta_1 & 0 \\ 0 & \Delta_2 \end{bmatrix} : \Delta_1 \in \Delta_1, \Delta_2 \in \Delta_2 \right\}. \quad (11.16)$$

Now, we may compute  $\mu$  with respect to three structures. The notations we use to keep track of these computations are as follows:  $\mu_1(\cdot)$  is with respect to  $\Delta_1$ ,  $\mu_2(\cdot)$  is with respect to  $\Delta_2$ , and  $\mu_\Delta(\cdot)$  is with respect to  $\Delta$ . In view of these notations,  $\mu_1(M_{11})$ ,  $\mu_2(M_{22})$  and  $\mu_\Delta(M)$  all make sense, though, for instance,  $\mu_1(M)$  does not.

This section is interested in following constant matrix problems:

- determine whether the LFT  $F_l(M, \Delta_2)$  is well posed for all  $\Delta_2 \in \Delta_2$  with  $\bar{\sigma}(\Delta_2) \leq \beta$  ( $< \beta$ ), and,
- if so, then determine how “large”  $F_l(M, \Delta_2)$  can get for this norm-bounded set of perturbations.

Let  $\Delta_2 \in \Delta_2$ . Recall that  $F_l(M, \Delta_2)$  is *well posed* if  $I - M_{22}\Delta_2$  is invertible. The first theorem is nothing more than a restatement of the definition of  $\mu$ .

**Theorem 11.6** *The linear fractional transformation  $F_l(M, \Delta_2)$  is well posed*

(a) for all  $\Delta_2 \in \mathbf{B}\Delta_2$  if and only if  $\mu_2(M_{22}) < 1$ .

(b) for all  $\Delta_2 \in \mathbf{B}^\circ\Delta_2$  if and only if  $\mu_2(M_{22}) \leq 1$ .

As the “perturbation”  $\Delta_2$  deviates from zero, the matrix  $F_l(M, \Delta_2)$  deviates from  $M_{11}$ . The range of values that  $\mu_1(F_l(M, \Delta_2))$  takes on is intimately related to  $\mu_\Delta(M)$ , as shown in the following theorem:

**Theorem 11.7 (MAIN LOOP THEOREM)** *The following are equivalent:*

$$\begin{aligned} \mu_\Delta(M) < 1 &\iff \begin{cases} \mu_2(M_{22}) < 1, \text{ and} \\ \max_{\Delta_2 \in \mathbf{B}\Delta_2} \mu_1(F_l(M, \Delta_2)) < 1. \end{cases} \\ \mu_\Delta(M) \leq 1 &\iff \begin{cases} \mu_2(M_{22}) \leq 1, \text{ and} \\ \max_{\Delta_2 \in \mathbf{B}^\circ\Delta_2} \mu_1(F_l(M, \Delta_2)) \leq 1. \end{cases} \end{aligned}$$

**Proof.** We shall only prove the first part of the equivalence. The proof for the second part is similar.

$\Leftarrow$  Let  $\Delta_i \in \Delta_i$  be given, with  $\bar{\sigma}(\Delta_i) \leq 1$ , and define  $\Delta = \text{diag} [\Delta_1, \Delta_2]$ . Obviously  $\Delta \in \Delta$ . Now

$$\det(I - M\Delta) = \det \begin{bmatrix} I - M_{11}\Delta_1 & -M_{12}\Delta_2 \\ -M_{21}\Delta_1 & I - M_{22}\Delta_2 \end{bmatrix}. \quad (11.17)$$

By hypothesis  $I - M_{22}\Delta_2$  is invertible, and hence,  $\det(I - M\Delta)$  becomes

$$\det(I - M_{22}\Delta_2) \det \left( I - M_{11}\Delta_1 - M_{12}\Delta_2 (I - M_{22}\Delta_2)^{-1} M_{21}\Delta_1 \right).$$

Collecting the  $\Delta_1$  terms leaves

$$\det(I - M\Delta) = \det(I - M_{22}\Delta_2) \det(I - F_l(M, \Delta_2)\Delta_1). \quad (11.18)$$

But,  $\mu_1(F_l(M, \Delta_2)) < 1$  and  $\Delta_1 \in \mathbf{B}\Delta_1$ , so  $I - F_l(M, \Delta_2)\Delta_1$  must be nonsingular. Therefore,  $I - M\Delta$  is nonsingular and, by definition,  $\mu_\Delta(M) < 1$ .

$\Rightarrow$  Basically, the argument above is reversed. Again let  $\Delta_1 \in \mathbf{B}\Delta_1$  and  $\Delta_2 \in \mathbf{B}\Delta_2$  be given, and define  $\Delta = \text{diag} [\Delta_1, \Delta_2]$ . Then  $\Delta \in \mathbf{B}\Delta$  and, by hypothesis,  $\det(I - M\Delta) \neq 0$ . It is easy to verify from the definition of  $\mu$  that (always)

$$\mu(M) \geq \max \{ \mu_1(M_{11}), \mu_2(M_{22}) \}.$$

We can see that  $\mu_2(M_{22}) < 1$ , which gives that  $I - M_{22}\Delta_2$  is also nonsingular. Therefore, the expression in (11.18) is valid, giving

$$\det(I - M_{22}\Delta_2) \det(I - F_l(M, \Delta_2)\Delta_1) = \det(I - M\Delta) \neq 0.$$

Obviously,  $I - F_l(M, \Delta_2) \Delta_1$  is nonsingular for all  $\Delta_i \in \mathbf{B}\Delta_i$ , which indicates that the claim is true.  $\square$

**Remark 11.3** This theorem forms the basis for all uses of  $\mu$  in linear system robustness analysis, whether from a state-space, frequency domain, or Lyapunov approach.  $\heartsuit$

The role of the block structure  $\Delta_2$  in the MAIN LOOP theorem is clear - it is the structure that the perturbations come from; however, the role of the perturbation structure  $\Delta_1$  is often misunderstood. Note that  $\mu_1(\cdot)$  appears on the right hand side of the theorem, so that the set  $\Delta_1$  defines what particular property of  $F_l(M, \Delta_2)$  is considered. As an example, consider the theorem applied with the two simple block structures considered right after Lemma 11.2. Define  $\Delta_1 := \{\delta_1 I_n : \delta_1 \in \mathbb{C}\}$ . Hence, for  $A \in \mathbb{C}^{n \times n}$ ,  $\mu_1(A) = \rho(A)$ . Likewise, define  $\Delta_2 = \mathbb{C}^{m \times m}$ ; then for  $D \in \mathbb{C}^{m \times m}$ ,  $\mu_2(D) = \bar{\sigma}(D)$ . Now, let  $\Delta$  be the diagonal augmentation of these two sets, namely

$$\Delta := \left\{ \begin{bmatrix} \delta_1 I_n & 0_{n \times m} \\ 0_{m \times n} & \Delta_2 \end{bmatrix} : \delta_1 \in \mathbb{C}, \Delta_2 \in \mathbb{C}^{m \times m} \right\} \subset \mathbb{C}^{(n+m) \times (n+m)}.$$

Let  $A \in \mathbb{C}^{n \times n}$ ,  $B \in \mathbb{C}^{n \times m}$ ,  $C \in \mathbb{C}^{m \times n}$ , and  $D \in \mathbb{C}^{m \times m}$  be given, and interpret them as the state space model of a discrete time system

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k \\ y_k &= Cx_k + Du_k. \end{aligned}$$

And let  $M \in \mathbb{C}^{(n+m) \times (n+m)}$  be the block state space matrix of the system

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

Applying the theorem with this data gives that the following are equivalent:

- The spectral radius of  $A$  satisfies  $\rho(A) < 1$ , and

$$\max_{\substack{\delta_1 \in \mathbb{C} \\ |\delta_1| \leq 1}} \bar{\sigma} \left( D + C \delta_1 (I - A \delta_1)^{-1} B \right) < 1. \quad (11.19)$$

- The maximum singular value of  $D$  satisfies  $\bar{\sigma}(D) < 1$ , and

$$\max_{\substack{\Delta_2 \in \mathbb{C}^{m \times m} \\ \bar{\sigma}(\Delta_2) \leq 1}} \rho \left( A + B \Delta_2 (I - D \Delta_2)^{-1} C \right) < 1. \quad (11.20)$$

- The structured singular value of  $M$  satisfies

$$\mu_{\Delta}(M) < 1. \quad (11.21)$$

The first condition is recognized by two things: the system is stable, and the  $\|\cdot\|_\infty$  norm on the transfer function from  $u$  to  $y$  is less than 1 (by replacing  $\delta_1$  with  $\frac{1}{z}$ )

$$\|G\|_\infty := \max_{\substack{z \in \mathbb{C} \\ |z| \geq 1}} \bar{\sigma} \left( D + C(zI - A)^{-1} B \right) = \max_{\substack{\delta_1 \in \mathbb{C} \\ |\delta_1| \leq 1}} \bar{\sigma} \left( D + C\delta_1 (I - A\delta_1)^{-1} B \right).$$

The second condition implies that  $(I - D\Delta_2)^{-1}$  is well defined for all  $\bar{\sigma}(\Delta_2) \leq 1$  and that a robust stability result holds for the uncertain difference equation

$$x_{k+1} = \left( A + B\Delta_2 (I - D\Delta_2)^{-1} C \right) x_k$$

where  $\Delta_2$  is any element in  $\mathbb{C}^{m \times m}$  with  $\bar{\sigma}(\Delta_2) \leq 1$ , but otherwise unknown.

This equivalence between the small gain condition,  $\|G\|_\infty < 1$ , and the stability robustness of the uncertain difference equation is well known. This is the small gain theorem, in its necessary and sufficient form for linear, time invariant systems with one of the components norm-bounded, but otherwise unknown. What is important to note is that both of these conditions are equivalent to a condition involving the structured singular value of the state space matrix. Already we have seen that special cases of  $\mu$  are the spectral radius and the maximum singular value. Here we see that other important linear system properties, namely robust stability and input-output gain, are also related to a particular case of the structured singular value.

## 11.3 Structured Robust Stability and Performance

### 11.3.1 Robust Stability

The most well-known use of  $\mu$  as a robustness analysis tool is in the frequency domain. Suppose  $G(s)$  is a stable, multi-input, multi-output transfer function of a linear system. For clarity, assume  $G$  has  $q_1$  inputs and  $p_1$  outputs. Let  $\Delta$  be a block structure, as in equation (11.3), and assume that the dimensions are such that  $\Delta \subset \mathbb{C}^{q_1 \times p_1}$ . We want to consider feedback perturbations to  $G$  which are themselves dynamical systems with the block-diagonal structure of the set  $\Delta$ .

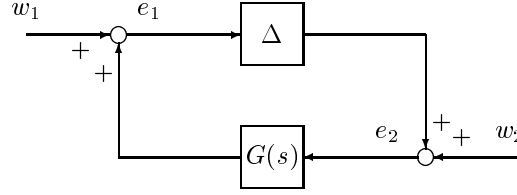
Let  $\mathcal{M}(\Delta)$  denote the set of all block diagonal and stable rational transfer functions that have block structures such as  $\Delta$ .

$$\mathcal{M}(\Delta) := \{ \Delta(\cdot) \in \mathcal{RH}_\infty : \Delta(s_o) \in \Delta \text{ for all } s_o \in \overline{\mathbb{C}}_+ \}$$

**Theorem 11.8** *Let  $\beta > 0$ . The loop shown below is well-posed and internally stable for all  $\Delta(\cdot) \in \mathcal{M}(\Delta)$  with  $\|\Delta\|_\infty < \frac{1}{\beta}$  if and only if*

$$\sup_{\omega \in \mathbb{R}} \mu_\Delta(G(j\omega)) \leq \beta$$





**Proof.** ( $\Leftarrow$ ) By Lemma 11.1,  $\sup_{s \in \overline{\mathbb{C}}_+} \mu_{\Delta}(G(s)) = \sup_{\omega \in \mathbb{R}} \mu_{\Delta}(G(j\omega)) \leq \beta$ . Hence  $\det(I - G(s)\Delta(s)) \neq 0$  for all  $s \in \overline{\mathbb{C}}_+ \cup \{\infty\}$  whenever  $\|\Delta\|_{\infty} < 1/\beta$ , i.e., the system is robustly stable.

( $\Rightarrow$ ) Suppose  $\sup_{\omega \in \mathbb{R}} \mu_{\Delta}(G(j\omega)) > \beta$ . Then there is a  $0 < \omega_o < \infty$  such that  $\mu_{\Delta}(G(j\omega_o)) > \beta$ . By Remark 11.1, there is a complex  $\Delta_c \in \Delta$  that each full block has rank 1 and  $\overline{\sigma}(\Delta_c) < 1/\beta$  such that  $I - G(j\omega_o)\Delta_c$  is singular. Next, using the same construction used in the proof of the small gain theorem (Theorem 9.1), one can find a rational  $\Delta(s)$  such that  $\|\Delta(s)\|_{\infty} = \overline{\sigma}(\Delta_c) < 1/\beta$ ,  $\Delta(j\omega_o) = \Delta_c$ , and  $\Delta(s)$  destabilizes the system.  $\square$

Hence, the peak value on the  $\mu$  plot of the frequency response determines the size of perturbations that the loop is robustly stable against.

**Remark 11.4** The internal stability with closed ball of uncertainties is more complicated. The following example is shown in Tits and Fan [1994]. Consider

$$G(s) = \frac{1}{s+1} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

and  $\Delta = \delta(s)I_2$ . Then

$$\sup_{\omega \in \mathbb{R}} \mu_{\Delta}(G(j\omega)) = \sup_{\omega \in \mathbb{R}} \frac{1}{|j\omega + 1|} = \mu_{\Delta}(G(j0)) = 1.$$

On the other hand,  $\mu_{\Delta}(G(s)) < 1$  for all  $s \neq 0, s \in \overline{\mathbb{C}}_+$ , and the only matrices in the form of  $\Gamma = \gamma I_2$  with  $|\gamma| \leq 1$  for which

$$\det(I - G(0)\Gamma) = 0$$

are the complex matrices  $\pm jI_2$ . Thus, clearly,  $(I - G(s)\Delta(s))^{-1} \in \mathcal{RH}_{\infty}$  for all real rational  $\Delta(s) = \delta(s)I_2$  with  $\|\delta\|_{\infty} \leq 1$  since  $\Delta(0)$  must be real. This shows that  $\sup_{\omega \in \mathbb{R}} \mu_{\Delta}(G(j\omega)) < 1$  is not necessary for  $(I - G(s)\Delta(s))^{-1} \in \mathcal{RH}_{\infty}$  with the closed ball of structured uncertainty  $\|\Delta\|_{\infty} \leq 1$ . Similar examples with no repeated blocks are generated by setting  $G(s) = \frac{1}{s+1}M$  where  $M$  is any real matrix with  $\mu_{\Delta}(M) = 1$  for which there is no real  $\Delta \in \Delta$  with  $\overline{\sigma}(\Delta) = 1$  such that  $\det(I - M\Delta) = 0$ . For example, let

$$M = \begin{bmatrix} 0 & \beta \\ \gamma & \alpha \\ \gamma & -\alpha \end{bmatrix} \begin{bmatrix} -\beta & \alpha & \alpha \\ 0 & -\gamma & \gamma \end{bmatrix}, \quad \Delta = \left\{ \begin{bmatrix} \delta_1 & & \\ & \delta_2 & \\ & & \delta_3 \end{bmatrix}, \delta_i \in \mathbb{C} \right\}$$

with  $\gamma^2 = \frac{1}{2}$  and  $\beta^2 + 2\alpha^2 = 1$ . Then it is shown in Packard and Doyle [1993] that  $\mu_{\Delta}(M) = 1$  and all  $\Delta \in \mathbf{\Delta}$  with  $\bar{\sigma}(\Delta) = 1$  that satisfy  $\det(I - M\Delta) = 0$  must be complex.  $\heartsuit$

**Remark 11.5** Let  $\Delta \in \mathcal{RH}_{\infty}$  be a structured uncertainty and

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \in \mathcal{RH}_{\infty}$$

then  $F_u(G, \Delta) \in \mathcal{RH}_{\infty}$  does not necessarily imply  $(I - G_{11}\Delta)^{-1} \in \mathcal{RH}_{\infty}$  whether  $\Delta$  is in an open ball or is in a closed ball. For example, consider

$$G(s) = \left[ \begin{array}{cc|c} \frac{1}{s+1} & 0 & 1 \\ 0 & \frac{10}{s+1} & 0 \\ \hline 1 & 0 & 0 \end{array} \right]$$

and  $\Delta = \begin{bmatrix} \delta_1 & \\ & \delta_2 \end{bmatrix}$  with  $\|\Delta\|_{\infty} < 1$ . Then  $F_u(G, \Delta) = \frac{1}{1 - \delta_1 \frac{1}{s+1}} \in \mathcal{RH}_{\infty}$  for all admissible  $\Delta$  ( $\|\Delta\|_{\infty} < 1$ ) but  $(I - G_{11}\Delta)^{-1} \in \mathcal{RH}_{\infty}$  is true only for  $\|\Delta\|_{\infty} < 0.1$ .  $\heartsuit$

### 11.3.2 Robust Performance

Often, stability is not the only property of a closed-loop system that must be robust to perturbations. Typically, there are exogenous disturbances acting on the system (wind gusts, sensor noise) which result in tracking and regulation errors. Under perturbation, the effect that these disturbances have on error signals can greatly increase. In most cases, long before the onset of instability, the closed-loop performance will degrade to the point of unacceptability, hence the need for a “robust performance” test. Such a test will indicate the worst-case level of performance degradation associated with a given level of perturbations.

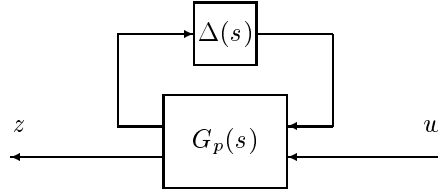
Assume  $G_p$  is a stable, real-rational, proper transfer function with  $q_1 + q_2$  inputs and  $p_1 + p_2$  outputs. Partition  $G_p$  in the obvious manner

$$G_p(s) = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$$

so that  $G_{11}$  has  $q_1$  inputs and  $p_1$  outputs, and so on. Let  $\mathbf{\Delta} \subset \mathbb{C}^{q_1 \times p_1}$  be a block structure, as in equation (11.3). Define an augmented block structure

$$\mathbf{\Delta}_P := \left\{ \begin{bmatrix} \Delta & 0 \\ 0 & \Delta_f \end{bmatrix} : \Delta \in \mathbf{\Delta}, \Delta_f \in \mathbb{C}^{q_2 \times p_2} \right\}.$$

The setup is to theoretically address the robust performance questions about the loop shown below



The transfer function from  $w$  to  $z$  is denoted by  $F_u(G_p, \Delta)$ .

**Theorem 11.9** *Let  $\beta > 0$ . For all  $\Delta(s) \in \mathcal{M}(\Delta)$  with  $\|\Delta\|_\infty < \frac{1}{\beta}$ , the loop shown above is well-posed, internally stable, and  $\|F_u(G_p, \Delta)\|_\infty \leq \beta$  if and only if*

$$\sup_{\omega \in \mathbb{R}} \mu_{\Delta_P}(G_p(j\omega)) \leq \beta.$$

Note that by internal stability,  $\sup_{\omega \in \mathbb{R}} \mu_{\Delta}(G_{11}(j\omega)) \leq \beta$ , then the proof of this theorem is exactly along the lines of the earlier proof for Theorem 11.8, but also appeals to Theorem 11.7. This is a remarkably useful theorem. It says that a robust performance problem is equivalent to a robust stability problem with augmented uncertainty  $\Delta$  as shown in Figure 11.5.

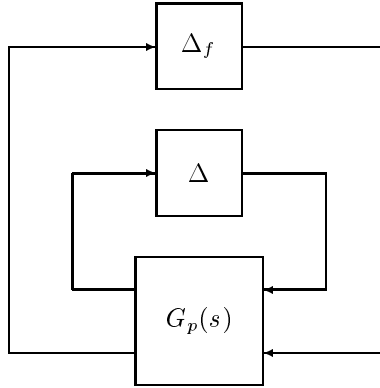


Figure 11.5: Robust Performance vs Robust Stability

### 11.3.3 Two Block $\mu$ : Robust Performance Revisited

Suppose that the uncertainty block is given by

$$\Delta = \begin{bmatrix} \Delta_1 & \\ & \Delta_2 \end{bmatrix} \in \mathcal{RH}_\infty$$

with  $\|\Delta\|_\infty < 1$  and that the interconnection model  $G$  is given by

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \in \mathcal{RH}_\infty.$$

Then the closed-loop system is well-posed and internally stable iff  $\sup_\omega \mu_\Delta(G(j\omega)) \leq 1$ . Let

$$D_\omega = \begin{bmatrix} d_\omega I & \\ & I \end{bmatrix}, \quad d_\omega \in \mathbb{R}_+$$

then

$$D_\omega G(j\omega) D_\omega^{-1} = \begin{bmatrix} G_{11}(j\omega) & d_\omega G_{12}(j\omega) \\ \frac{1}{d_\omega} G_{21}(j\omega) & G_{22}(j\omega) \end{bmatrix}.$$

Hence by Theorem 11.5, at each frequency  $\omega$

$$\mu_\Delta(G(j\omega)) = \inf_{d_\omega \in \mathbb{R}_+} \bar{\sigma} \left( \begin{bmatrix} G_{11}(j\omega) & d_\omega G_{12}(j\omega) \\ \frac{1}{d_\omega} G_{21}(j\omega) & G_{22}(j\omega) \end{bmatrix} \right). \quad (11.22)$$

Since the minimization is convex in  $\log d_\omega$  [see, Doyle, 1982], the optimal  $d_\omega$  can be found by a search; however, two approximations to  $d_\omega$  can be obtained easily by approximating the right hand side of (11.22):

(1) From Lemma 2.10, we have

$$\begin{aligned} \mu_\Delta(G(j\omega)) &\leq \inf_{d_\omega \in \mathbb{R}_+} \bar{\sigma} \left( \begin{bmatrix} \|G_{11}(j\omega)\| & d_\omega \|G_{12}(j\omega)\| \\ \frac{1}{d_\omega} \|G_{21}(j\omega)\| & \|G_{22}(j\omega)\| \end{bmatrix} \right) \\ &\leq \sqrt{\inf_{d_\omega \in \mathbb{R}_+} \left( \|G_{11}(j\omega)\|^2 + d_\omega^2 \|G_{12}(j\omega)\|^2 + \frac{1}{d_\omega^2} \|G_{21}(j\omega)\|^2 + \|G_{22}(j\omega)\|^2 \right)} \\ &= \sqrt{\|G_{11}(j\omega)\|^2 + \|G_{22}(j\omega)\|^2 + 2 \|G_{12}(j\omega)\| \|G_{21}(j\omega)\|} \end{aligned}$$

with the minimizing  $d_\omega$  given by

$$\hat{d}_\omega = \begin{cases} \sqrt{\frac{\|G_{21}(j\omega)\|}{\|G_{12}(j\omega)\|}} & \text{if } G_{12} \neq 0 \text{ \& } G_{21} \neq 0, \\ 0 & \text{if } G_{21} = 0, \\ \infty & \text{if } G_{12} = 0. \end{cases} \quad (11.23)$$

(2) Alternative approximation can be obtained by using the Frobenius norm

$$\mu_\Delta(G(j\omega)) \leq \inf_{d_\omega \in \mathbb{R}_+} \left\| \begin{bmatrix} G_{11}(j\omega) & d_\omega G_{12}(j\omega) \\ \frac{1}{d_\omega} G_{21}(j\omega) & G_{22}(j\omega) \end{bmatrix} \right\|_F$$

$$\begin{aligned}
&= \sqrt{\inf_{d_\omega \in \mathbb{R}_+} \left( \|G_{11}(j\omega)\|_F^2 + d_\omega^2 \|G_{12}(j\omega)\|_F^2 + \frac{1}{d_\omega^2} \|G_{21}(j\omega)\|_F^2 + \|G_{22}(j\omega)\|_F^2 \right)} \\
&= \sqrt{\|G_{11}(j\omega)\|_F^2 + \|G_{22}(j\omega)\|_F^2 + 2 \|G_{12}(j\omega)\|_F \|G_{21}(j\omega)\|_F}
\end{aligned}$$

with the minimizing  $d_\omega$  given by

$$\tilde{d}_\omega = \begin{cases} \sqrt{\frac{\|G_{21}(j\omega)\|_F}{\|G_{12}(j\omega)\|_F}} & \text{if } G_{12} \neq 0 \text{ \& } G_{21} \neq 0, \\ 0 & \text{if } G_{21} = 0, \\ \infty & \text{if } G_{12} = 0. \end{cases} \quad (11.24)$$

It can be shown that the approximations for the scalar  $d_\omega$  obtained above are exact for a  $2 \times 2$  matrix  $G$ . For higher dimensional  $G$ , the approximations for  $d_\omega$  are still reasonably good. Hence an approximation of  $\mu$  can be obtained as

$$\mu_\Delta(G(j\omega)) \leq \bar{\sigma} \left( \begin{bmatrix} G_{11}(j\omega) & \hat{d}_\omega G_{12}(j\omega) \\ \frac{1}{\hat{d}_\omega} G_{21}(j\omega) & G_{22}(j\omega) \end{bmatrix} \right) \quad (11.25)$$

or, alternatively, as

$$\mu_\Delta(G(j\omega)) \leq \bar{\sigma} \left( \begin{bmatrix} G_{11}(j\omega) & \tilde{d}_\omega G_{12}(j\omega) \\ \frac{1}{\tilde{d}_\omega} G_{21}(j\omega) & G_{22}(j\omega) \end{bmatrix} \right). \quad (11.26)$$

We can now see how these approximated  $\mu$  tests are compared with the sufficient conditions obtained in Chapter 9.

**Example 11.1** Consider again the robust performance problem of a system with output multiplicative uncertainty in Chapter 9 (see Figure 9.5):

$$P_\Delta = (I + W_1 \Delta W_2) P, \quad \|\Delta\|_\infty < 1.$$

Then it is easy to show that the problem can be put in the general framework by selecting

$$G(s) = \begin{bmatrix} -W_2 T_o W_1 & -W_2 T_o W_d \\ W_e S_o W_1 & W_e S_o W_d \end{bmatrix}$$

and that the robust performance condition is satisfied if and only if

$$\|W_2 T_o W_1\|_\infty \leq 1 \quad (11.27)$$

and

$$\|\mathcal{F}_u(G, \Delta)\|_\infty \leq 1 \quad (11.28)$$

for all  $\Delta \in \mathcal{RH}_\infty$  with  $\|\Delta\|_\infty < 1$ . But (11.27) and (11.28) are satisfied iff for each frequency  $\omega$

$$\mu_\Delta(G(j\omega)) = \inf_{d_\omega \in \mathbb{R}_+} \bar{\sigma} \left( \begin{bmatrix} -W_2 T_o W_1 & -d_\omega W_2 T_o W_d \\ \frac{1}{d_\omega} W_e S_o W_1 & W_e S_o W_d \end{bmatrix} \right) \leq 1.$$

Note that, in contrast to the sufficient condition obtained in Chapter 9, this condition is an exact test for robust performance. To compare the  $\mu$  test with the criteria obtained in Chapter 9, some upper bounds for  $\mu$  can be derived. Let

$$d_\omega = \sqrt{\frac{\|W_e S_o W_1\|}{\|W_2 T_o W_d\|}}.$$

Then, using the first approximation for  $\mu$ , we get

$$\begin{aligned} \mu_\Delta(G(j\omega)) &\leq \sqrt{\|W_2 T_o W_1\|^2 + \|W_e S_o W_d\|^2 + 2 \|W_2 T_o W_d\| \|W_e S_o W_1\|} \\ &\leq \sqrt{\|W_2 T_o W_1\|^2 + \|W_e S_o W_d\|^2 + 2\kappa(W_1^{-1} W_d) \|W_2 T_o W_1\| \|W_e S_o W_d\|} \\ &\leq \|W_2 T_o W_1\| + \kappa(W_1^{-1} W_d) \|W_e S_o W_d\| \end{aligned}$$

where  $W_1$  is assumed to be invertible in the last two inequalities. The last term is exactly the sufficient robust performance criteria obtained in Chapter 9. It is clear that any term preceding the last forms a tighter test since  $\kappa(W_1^{-1} W_d) \geq 1$ . Yet another alternative sufficient test can be obtained from the above sequence of inequalities:

$$\mu_\Delta(G(j\omega)) \leq \sqrt{\kappa(W_1^{-1} W_d) (\|W_2 T_o W_1\| + \|W_e S_o W_d\|)}.$$

Note that this sufficient condition is not easy to get from the approach taken in Chapter 9 and is potentially less conservative than the bounds derived there.  $\diamond$

Next we consider the skewed specification problem, but first the following lemma is needed in the sequel.

**Lemma 11.10** *Suppose  $\bar{\sigma} = \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m = \underline{\sigma} > 0$ , then*

$$\inf_{d \in \mathbb{R}_+} \max_i \left\{ (d\sigma_i)^2 + \frac{1}{(d\sigma_i)^2} \right\} = \frac{\bar{\sigma}}{\underline{\sigma}} + \frac{\underline{\sigma}}{\bar{\sigma}}.$$

**Proof.** Consider a function  $y = x + 1/x$ , then  $y$  is a convex function and the maximization over a closed interval is achieved at the boundary of the interval. Hence for any fixed  $d$

$$\max_i \left\{ (d\sigma_i)^2 + \frac{1}{(d\sigma_i)^2} \right\} = \max \left\{ (d\bar{\sigma})^2 + \frac{1}{(d\bar{\sigma})^2}, (d\underline{\sigma})^2 + \frac{1}{(d\underline{\sigma})^2} \right\}.$$

Then the minimization over  $d$  is obtained iff

$$(d\bar{\sigma})^2 + \frac{1}{(d\bar{\sigma})^2} = (d\underline{\sigma})^2 + \frac{1}{(d\underline{\sigma})^2}$$

which gives  $d^2 = \frac{1}{\bar{\sigma}\underline{\sigma}}$ . The result then follows from substituting  $d$ .  $\square$

**Example 11.2** As another example, consider again the skewed specification problem from Chapter 9. Then the corresponding  $G$  matrix is given by

$$G = \begin{bmatrix} -W_2 T_i W_1 & -W_2 K S_o W_d \\ W_e S_o P W_1 & W_e S_o W_d \end{bmatrix}.$$

So the robust performance specification is satisfied iff

$$\mu_{\Delta}(G(j\omega)) = \inf_{d_\omega \in \mathbb{R}_+} \bar{\sigma} \left( \begin{bmatrix} -W_2 T_i W_1 & -d_\omega W_2 K S_o W_d \\ \frac{1}{d_\omega} W_e S_o P W_1 & W_e S_o W_d \end{bmatrix} \right) \leq 1$$

for all  $\omega \geq 0$ . As in the last example, an upper bound can be obtained by taking

$$d_\omega = \sqrt{\frac{\|W_e S_o P W_1\|}{\|W_2 K S_o W_d\|}}.$$

Then

$$\mu_{\Delta}(G(j\omega)) \leq \sqrt{\kappa(W_d^{-1} P W_1)(\|W_2 T_i W_1\| + \|W_e S_o W_d\|)}.$$

In particular, this suggests that the robust performance margin is inversely proportional to the square root of the plant condition number if  $W_d = I$  and  $W_1 = I$ . This can be further illustrated by considering a plant-inverting control system.

To simplify the exposition, we shall make the following assumptions:

$$W_e = w_s I, \quad W_d = I, \quad W_1 = I, \quad W_2 = w_t I,$$

and  $P$  is stable and has a stable inverse (i.e., minimum phase) ( $P$  can be strictly proper). Furthermore, we shall assume that the controller has the form

$$K(s) = P^{-1}(s)l(s)$$

where  $l(s)$  is a scalar loop transfer function which makes  $K(s)$  proper and stabilizes the closed-loop. This compensator produces diagonal sensitivity and complementary sensitivity functions with identical diagonal elements, namely

$$S_o = S_i = \frac{1}{1 + l(s)} I, \quad T_o = T_i = \frac{l(s)}{1 + l(s)} I.$$

Denote

$$\varepsilon(s) = \frac{1}{1+l(s)}, \quad \tau(s) = \frac{l(s)}{1+l(s)}$$

and substitute these expressions into  $G$ ; we get

$$G = \begin{bmatrix} -w_t \tau I & -w_t \tau P^{-1} \\ w_s \varepsilon P & w_s \varepsilon I \end{bmatrix}.$$

The structured singular value for  $G$  at frequency  $\omega$  can be computed by

$$\mu_{\Delta}(G(j\omega)) = \inf_{d \in \mathbb{R}_+} \bar{\sigma} \left( \begin{bmatrix} -w_t \tau I & -w_t \tau (dP)^{-1} \\ w_s \varepsilon dP & w_s \varepsilon I \end{bmatrix} \right).$$

Let the singular value decomposition of  $P(j\omega)$  at frequency  $\omega$  be

$$P(j\omega) = U \Sigma V^*, \quad \Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_m)$$

with  $\sigma_1 = \bar{\sigma}$  and  $\sigma_m = \underline{\sigma}$  where  $m$  is the dimension of  $P$ . Then

$$\mu_{\Delta}(G(j\omega)) = \inf_{d \in \mathbb{R}_+} \bar{\sigma} \left( \begin{bmatrix} -w_t \tau I & -w_t \tau (d\Sigma)^{-1} \\ w_s \varepsilon d\Sigma & w_s \varepsilon I \end{bmatrix} \right)$$

since unitary operations do not change the singular values of a matrix. Note that

$$\begin{bmatrix} -w_t \tau I & -w_t \tau (d\Sigma)^{-1} \\ w_s \varepsilon d\Sigma & w_s \varepsilon I \end{bmatrix} = P_1 \text{diag}(M_1, M_2, \dots, M_m) P_2$$

where  $P_1$  and  $P_2$  are permutation matrices and where

$$M_i = \begin{bmatrix} -w_t \tau & -w_t \tau (d\sigma_i)^{-1} \\ w_s \varepsilon d\sigma_i & w_s \varepsilon \end{bmatrix}.$$

Hence

$$\begin{aligned} \mu_{\Delta}(G(j\omega)) &= \inf_{d \in \mathbb{R}_+} \max_i \bar{\sigma} \left( \begin{bmatrix} -w_t \tau & -w_t \tau (d\sigma_i)^{-1} \\ w_s \varepsilon d\sigma_i & w_s \varepsilon \end{bmatrix} \right) \\ &= \inf_{d \in \mathbb{R}_+} \max_i \bar{\sigma} \left( \begin{bmatrix} -w_t \tau \\ w_s \varepsilon d\sigma_i \end{bmatrix} \begin{bmatrix} 1 & (d\sigma_i)^{-1} \end{bmatrix} \right) \\ &= \inf_{d \in \mathbb{R}_+} \max_i \sqrt{(1 + |d\sigma_i|^{-2})(|w_s \varepsilon d\sigma_i|^2 + |w_t \tau|^2)} \\ &= \inf_{d \in \mathbb{R}_+} \max_i \sqrt{|w_s \varepsilon|^2 + |w_t \tau|^2 + |w_s \varepsilon d\sigma_i|^2 + \left| \frac{w_t \tau}{d\sigma_i} \right|^2}. \end{aligned}$$



Using Lemma 11.10, it is easy to show that the maximum is achieved at either  $\bar{\sigma}$  or  $\underline{\sigma}$  and that optimal  $d$  is given by

$$d^2 = \frac{|w_t \tau|}{|w_s \varepsilon| \underline{\sigma} \bar{\sigma}},$$

so the structured singular value is

$$\mu_{\Delta}(G(j\omega)) = \sqrt{|w_s \varepsilon|^2 + |w_t \tau|^2 + |w_s \varepsilon| |w_t \tau| [\kappa(P) + \frac{1}{\kappa(P)}]}. \quad (11.29)$$

Note that if  $|w_s \varepsilon|$  and  $|w_t \tau|$  are not too large, which are guaranteed if the nominal performance and robust stability conditions are satisfied, then the structured singular value is proportional to the square root of the plant condition number:

$$\mu_{\Delta}(G(j\omega)) \approx \sqrt{|w_s \varepsilon| |w_t \tau| \kappa(P)}. \quad (11.30)$$

◇

This confirms our intuition that an ill-conditioned plant with skewed specifications is hard to control.

#### 11.3.4 Approximation of Multiple Full Block $\mu$

The approximations given in the last subsection can be generalized to the multiple block  $\mu$  problem by assuming that  $M$  is partitioned consistently with the structure of

$$\Delta = \text{diag}(\Delta_1, \Delta_2, \dots, \Delta_F)$$

so that

$$M = \begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1F} \\ M_{21} & M_{22} & \cdots & M_{2F} \\ \vdots & \vdots & & \vdots \\ M_{F1} & M_{F2} & \cdots & M_{FF} \end{bmatrix}$$

and

$$D = \text{diag}(d_1 I, \dots, d_{F-1} I, I).$$

Now

$$DMD^{-1} = \left[ M_{ij} \frac{d_i}{d_j} \right], \quad d_F := 1.$$

And hence

$$\mu_{\Delta}(M) \leq \inf_{D \in \mathcal{D}} \bar{\sigma}(DMD^{-1}) = \inf_{D \in \mathcal{D}} \bar{\sigma} \left[ M_{ij} \frac{d_i}{d_j} \right]$$

$$\begin{aligned}
&\leq \inf_{D \in \mathcal{D}} \bar{\sigma} \left[ \|M_{ij}\| \frac{d_i}{d_j} \right] \leq \inf_{D \in \mathcal{D}} \sqrt{\sum_{i=1}^F \sum_{j=1}^F \|M_{ij}\|^2 \frac{d_i^2}{d_j^2}} \\
&\leq \inf_{D \in \mathcal{D}} \sqrt{\sum_{i=1}^F \sum_{j=1}^F \|M_{ij}\|_F^2 \frac{d_i^2}{d_j^2}}.
\end{aligned}$$

An approximate  $D$  can be found by solving the following minimization problem:

$$\inf_{D \in \mathcal{D}} \sum_{i=1}^F \sum_{j=1}^F \|M_{ij}\|^2 \frac{d_i^2}{d_j^2}$$

or, more conveniently, by minimizing

$$\inf_{D \in \mathcal{D}} \sum_{i=1}^F \sum_{j=1}^F \|M_{ij}\|_F^2 \frac{d_i^2}{d_j^2}$$

with  $d_F = 1$ . The optimal  $d_i$  minimizing of the above two problems satisfy, respectively,

$$d_k^4 = \frac{\sum_{i \neq k} \|M_{ik}\|^2 d_i^2}{\sum_{j \neq k} \|M_{kj}\|^2 / d_j^2}, \quad k = 1, 2, \dots, F-1 \quad (11.31)$$

and

$$d_k^4 = \frac{\sum_{i \neq k} \|M_{ik}\|_F^2 d_i^2}{\sum_{j \neq k} \|M_{kj}\|_F^2 / d_j^2}, \quad k = 1, 2, \dots, F-1. \quad (11.32)$$

Using these relations,  $d_k$  can be obtained by iterations.

**Example 11.3** Consider a  $3 \times 3$  complex matrix

$$M = \begin{bmatrix} 1+j & 10-2j & -20j \\ 5j & 3+j & -1+3j \\ -2 & j & 4-j \end{bmatrix}$$

with structured  $\Delta = \text{diag}(\delta_1, \delta_2, \delta_3)$ . The largest singular value of  $M$  is  $\bar{\sigma}(M) = 22.9094$  and the structured singular value of  $M$  computed using the  $\mu$ -TOOLS is equal to its upper bound:

$$\mu_{\Delta}(M) = \inf_{D \in \mathcal{D}} \bar{\sigma}(DM D^{-1}) = 11.9636$$

with the optimal scaling  $D_{opt} = \text{diag}(0.3955, 0.6847, 1)$ . The optimal  $D$  minimizing

$$\inf_{D \in \mathcal{D}} \sum_{i=1}^F \sum_{j=1}^F \|M_{ij}\|^2 \frac{d_i^2}{d_j^2}$$

is  $D_{subopt} = \text{diag}(0.3212, 0.4643, 1)$  which is solved from equation (11.31). Using this  $D_{subopt}$ , we obtain another upper bound for the structured singular value:

$$\mu_{\Delta}(M) \leq \bar{\sigma}(D_{subopt} M D_{subopt}^{-1}) = 12.2538.$$

One may also use this  $D_{subopt}$  as an initial guess for the exact optimization.  $\diamond$

## 11.4 Overview on $\mu$ Synthesis

This section briefly outlines various synthesis methods. The details are somewhat complicated and are treated in the other parts of the book. At this point, we simply want to point out how the analysis theory discussed in the previous sections leads naturally to synthesis questions.

From the analysis results, we see that each case eventually leads to the evaluation of

$$\|M\|_{\alpha} \quad \alpha = 2, \infty, \text{ or } \mu \quad (11.33)$$

for some transfer matrix  $M$ . Thus when the controller is put back into the problem, it involves only a simple linear fractional transformation as shown in Figure 11.6 with

$$M = \mathcal{F}_{\ell}(G, K) = G_{11} + G_{12}K(I - G_{22}K)^{-1}G_{21}.$$

where  $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$  is chosen, respectively, as

- nominal performance only ( $\Delta = 0$ ):  $G = \begin{bmatrix} P_{22} & P_{23} \\ P_{32} & P_{33} \end{bmatrix}$
- robust stability only:  $G = \begin{bmatrix} P_{11} & P_{13} \\ P_{31} & P_{33} \end{bmatrix}$
- robust performance:  $G = P = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix}$ .

Each case then leads to the synthesis problem

$$\min_K \|\mathcal{F}_{\ell}(G, K)\|_{\alpha} \quad \text{for } \alpha = 2, \infty, \text{ or } \mu \quad (11.34)$$

which is subject to the internal stability of the nominal.

The solutions of these problems for  $\alpha = 2$  and  $\infty$  are the focus of the rest of this book. The solutions presented in this book unify the two approaches in a common synthesis framework. The  $\alpha = 2$  case was already known in the 1960's, and the results

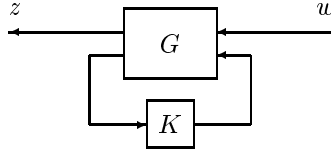


Figure 11.6: Synthesis Framework

are simply a new interpretation. The two Riccati solutions for the  $\alpha = \infty$  case were new products of the late 1980's.

The synthesis for the  $\alpha = \mu$  case is not yet fully solved. Recall that  $\mu$  may be obtained by scaling and applying  $\|\cdot\|_\infty$  (for  $F \leq 3$  and  $S = 0$ ), a reasonable approach is to “solve”

$$\min_K \inf_{D, D^{-1} \in \mathcal{H}_\infty} \|D\mathcal{F}_\ell(G, K)D^{-1}\|_\infty \quad (11.35)$$

by iteratively solving for  $K$  and  $D$ . This is the so-called *D-K Iteration*. The stable and minimum phase scaling matrix  $D(s)$  is chosen such that  $D(s)\Delta(s) = \Delta(s)D(s)$  (Note that  $D(s)$  is not necessarily belong to  $\mathcal{D}$  since  $D(s)$  is not necessarily Hermitian, see Remark 11.2). For a fixed scaling transfer matrix  $D$ ,  $\min_K \|D\mathcal{F}_\ell(G, K)D^{-1}\|_\infty$  is a standard  $\mathcal{H}_\infty$  optimization problem which will be solved in the later part of the book. For a given stabilizing controller  $K$ ,  $\inf_{D, D^{-1} \in \mathcal{H}_\infty} \|D\mathcal{F}_\ell(G, K)D^{-1}\|_\infty$  is a standard convex optimization problem and it can be solved pointwise in the frequency domain:

$$\sup_\omega \inf_{D_\omega \in \mathcal{D}} \bar{\sigma} [D_\omega \mathcal{F}_\ell(G, K)(j\omega)D_\omega^{-1}].$$

Indeed,

$$\inf_{D, D^{-1} \in \mathcal{H}_\infty} \|D\mathcal{F}_\ell(G, K)D^{-1}\|_\infty = \sup_\omega \inf_{D_\omega \in \mathcal{D}} \bar{\sigma} [D_\omega \mathcal{F}_\ell(G, K)(j\omega)D_\omega^{-1}].$$

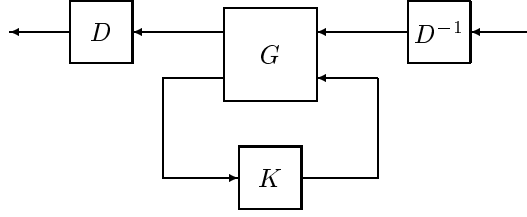
This follows intuitively from the following arguments: the left hand side is always no smaller than the right hand side, and, on the other hand, given the minimizing  $D_\omega$  from the right hand side across the frequency, there is always a rational function  $D(s)$  uniformly approximating the magnitude frequency response  $D_\omega$ .

Note that when  $S = 0$ , (no scalar blocks)

$$D_\omega = \text{diag}(d_1^\omega I, \dots, d_{F-1}^\omega I, I) \in \mathcal{D},$$

which is a block-diagonal scaling matrix applied pointwise across frequency to the frequency response  $\mathcal{F}_\ell(G, K)(j\omega)$ .

D-K Iterations proceed by performing this two-parameter minimization in sequential fashion: first minimizing over  $K$  with  $D_\omega$  fixed, then minimizing pointwise over  $D_\omega$  with  $K$  fixed, then again over  $K$ , and again over  $D_\omega$ , etc. Details of this process are summarized in the following steps:

Figure 11.7:  $\mu$ -Synthesis via Scaling

- (i) Fix an initial estimate of the scaling matrix  $D_\omega \in \mathcal{D}$  pointwise across frequency.
- (ii) Find scalar transfer functions  $d_i(s), d_i^{-1}(s) \in \mathcal{RH}_\infty$  for  $i = 1, 2, \dots, (F - 1)$  such that  $|d_i(j\omega)| \approx d_i^\omega$ . This step can be done using the interpolation theory [Youla and Saito, 1967]; however, this will usually result in very high order transfer functions, which explains why this process is currently done mostly by graphical matching using lower order transfer functions.
- (iii) Let

$$D(s) = \text{diag}(d_1(s)I, \dots, d_{F-1}(s)I, I).$$

Construct a state space model for system

$$\hat{G}(s) = \begin{bmatrix} D(s) & \\ & I \end{bmatrix} G(s) \begin{bmatrix} D^{-1}(s) & \\ & I \end{bmatrix}.$$

- (iv) Solve an  $\mathcal{H}_\infty$ -optimization problem to minimize

$$\left\| \mathcal{F}_\ell(\hat{G}, K) \right\|_\infty$$

over all stabilizing  $K$ 's. Note that this optimization problem uses the scaled version of  $G$ . Let its minimizing controller be denoted by  $\hat{K}$ .

- (v) Minimize  $\bar{\sigma}[D_\omega \mathcal{F}_\ell(G, \hat{K}) D_\omega^{-1}]$  over  $D_\omega$ , pointwise across frequency.<sup>4</sup> Note that this evaluation uses the minimizing  $\hat{K}$  from the last step, but that  $G$  is unscaled. The minimization itself produces a new scaling function. Let this new function be denoted by  $\hat{D}_\omega$ .
- (vi) Compare  $\hat{D}_\omega$  with the previous estimate  $D_\omega$ . Stop if they are close, but, otherwise, replace  $D_\omega$  with  $\hat{D}_\omega$  and return to step (ii).

With either  $K$  or  $D$  fixed, the global optimum in the other variable may be found using the  $\mu$  and  $\mathcal{H}_\infty$  solutions. Although the joint optimization of  $D$  and  $K$  is not convex and

<sup>4</sup>The approximate solutions given in the last section may be used.

the global convergence is not guaranteed, many designs have shown that this approach works very well [see e.g. Balas, 1990]. In fact, this is probably the most effective design methodology available today for dealing with such complicated problems. The detailed treatment of  $\mu$  analysis is given in Packard and Doyle [1991]. The rest of this book will focus on the  $\mathcal{H}_\infty$  optimization which is a fundamental tool for  $\mu$  synthesis.

## 11.5 Notes and References

This chapter is partially based on the lecture notes given by Doyle [1984] in Honeywell and partially based on the lecture notes by Packard [1991] and the paper by Doyle, Packard, and Zhou [1991]. Parts of section 11.3.3 come from the paper by Stein and Doyle [1991]. The small  $\mu$  theorem for systems with non-rational plants and uncertainties is proven in Tits [1994]. Other results on  $\mu$  can be found in Fan and Tits [1986], Fan, Tits, and Doyle [1991], Packard and Doyle [1993], Packard and Pandey [1993], Young [1993], and references therein.